MIL-E-1/1698A(EC)
15 April 1976
SUPERSEDING
MIL-E-1/1698(NAVY)
13 July 1970

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RADIATION COUNTER

TYPE 7301

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Low sensitivity, self-quenching for detection of gamma radiation

See figure 1

Mounting position: Any

Weight: 0.45 ounces (12.76 grams)

ABSOLUTE RATINGS:

Parameter:	Ebb	Rp	TA
Unit:	Vdc	Meg	°C
Maximum:	725	---	+75
Minimum:	675	---	-40
TEST CONDITIONS:	700	1.0	---

GENERAL:

Qualification - Required
Marking - See note 1

Ⓐ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 1								
6201	Background, contamination, and photosensitivity	2,3,4	t = 2 minutes	0.65	II	N/t	---	10	Npm
6211	Starting voltage	4	Pulse amplitude = 1 V; Nps = 200 (max); R2 = 1 Meg; R1 = C1 = C2 = 0; C3 = 0.01 µF	0.65	II	Es	---	640	Vdc
6216	Relative plateau slope	5	Voltage range = 660 to 760 Vdc	0.65	II	Ps	---	0.2	%/Vdc
6216	End-of-plateau voltage	5	Ps = 0.3/Vdc	0.65	II	Ee	780	---	Vdc
6221	Response count rate and current (gamma)	5,6,7	t = 2 minutes	0.65	II	N/t	75	125	Nps
	Quality conformance inspection, part 2								
6221	Response current	-	Gamma (radium) rate = 500 mr/hr; Ebb = 710 Vdc	---	---	Ib	---	10.0	µAdc
6226	Pulse amplitude (1)	4	Ebb = 675 Vdc; Nps = 200 (max); R1 = 0.9 Meg; R2 = 0.1 Meg; C1 = 330 pF; C2 = 3,000 pF; C3 = 0.01 µF; multiply oscilloscope reading by 10	---	---	eo	2.5	---	v
6226	Pulse amplitude (2)	4	Pulse amplitude (1), except Ebb = 725 Vdc	---	---	eo	---	65	v
	Quality conformance inspection, part 3								
---	Life test	-	Group A; counting rate = 1,000 Nps (min); t = 500 hours	---	---	---	---	---	---
---	Life-test end points	10		---	---	---	---	---	---
1031	Variable-frequency vibration	8,11, 13		---	---	---	---	---	---
---	Temperature cycling	12,13		---	---	---	---	---	---
(A)1041	Shock	8,9,13	600 G	---	---	---	---	---	---
6205	Leakage current	13	Ebb = 500 Vdc	---	---	LIb	---	0.5	µAdc

NOTES:

1. Each tube shall have an individual serial number permanently marked on the cathode.
2. In determining tube response (count rate) Ebb = 700 Vdc and Rp = 1 Meg. Tube count rates shall be determined using a scaler having a resolving time of 5 microseconds and a discrimination level of 0.25 volts.
3. The tube shall be exposed to radiation from a General Electric 15-watt germicidal lamp and a General Electric 15-watt fluorescent lamp, or equivalents, with relative positions as shown on figure 2.
4. Test shall be performed at the conclusion of the holding period.
5. Method B (test method 6216); the scaler of note 2 may be utilized for this test.

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NOTES: -Continued

6. The response characteristics, current and count rate versus field intensity of a typical 7301 radiation counter tube shall be as shown on figure 3. In determining tube response, Ebb = 700 Vdc and Rp = 1 Meg. Tube current shown as measured by means of a microammeter in series with the tube cathode, and the count rates as determined using a scaler having a resolving time of 5 microseconds and a discrimination level of 0.25 volts.

7. Use government approved standard gamma test unit.

8. Criterion for passing this test shall be compliance after test of at least 80 percent of the tubes with the requirements for:

   (a) Starting voltage.
   (b) Gamma response (count rate).
   (c) Pulse amplitude (1) and (2).
   (d) End-of-plateau voltage.
   (e) Relative plateau slope.

9. The tube shall be rigidly mounted in positions X, Y, and Z, by means of a clamp at the center of the cathode. The tube shall be given 10 blows at the X position and 5 at each of the other test positions.

10. The limits for acceptability shall be as follows:

    Background, contamination and photosensitivity: 12 Npm, maximum.
    Starting voltage: initial value ±10 volts.
    Plateau length: 125 volts, minimum.
    Relative plateau slope: 0.25 percent/Vdc, maximum.
    End-of-plateau voltage: 750 volts, minimum.
    Gamma response (count rate): initial value -10 percent.
    Pulse amplitude (1) and (2): initial limits.

11. The tube shall be mounted with its longitudinal axis horizontal and perpendicular to the direction of motion by means of a rigid clamp at the center of the cathode. The time for gradually covering the frequency range shall be 15 minutes.

12. With the tube in a field giving 100 ± 10 Nps at 700 Vdc, the tube response (count rate) at 675, 700, and 725 Vdc shall be determined at each of the following temperatures and in the order shown.

    1. Room temperature.
    2. -40°C.
    3. Room temperature.
    4. 75°C.
    5. Room temperature.

    A minimum stabilization time of 30 minutes shall be allowed at each temperature. The absolute count rate at 700 Vdc, and the average relative plateau slope as determined by the readings at the 3 voltages (best average position of straight line), shall not differ from the initial readings at room temperature by more than 10 percent and ±0.1 percent/volt, respectively, at any of the four subsequent temperatures shown.

(A) 13. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

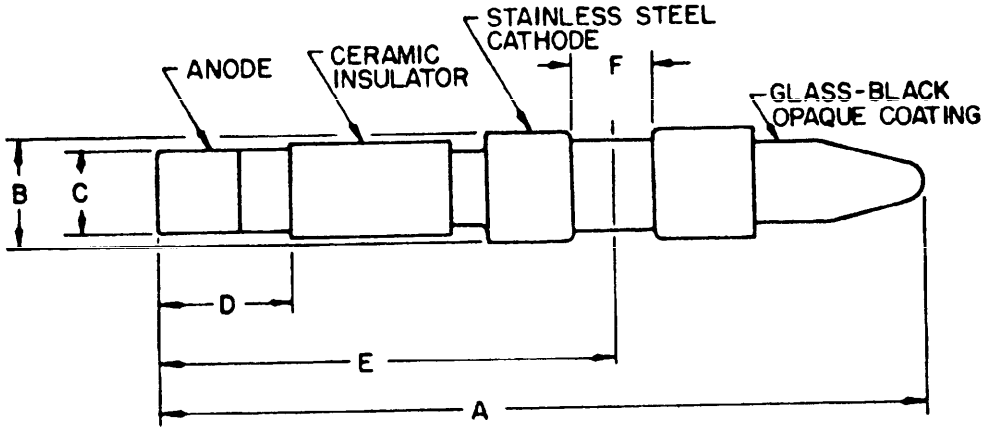
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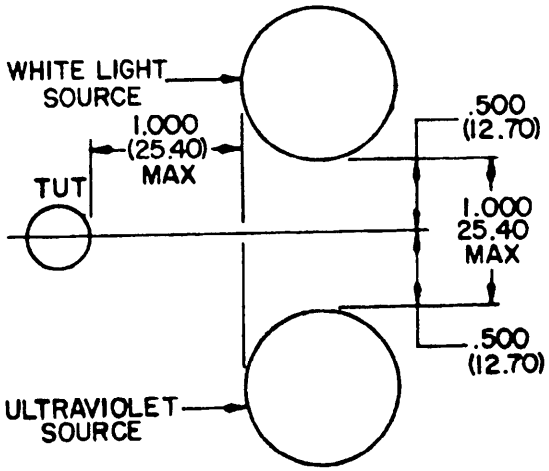
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A		2.375 (60.32)
B	.334 (8.48)	.354 (8.99)
C	.245 (6.22)	.255 (6.48)
D	.250 (6.35)	
E	1.344 (34.14)	1.468 (37.29)
F	.250 (6.35)	

NOTE:
Provision must be made to hold the anode wire concentric to the cathode at each end of the cathode. Anode wire is continuously maintained in tension.

Ⓐ FIGURE 1. Outline drawing of electron tube type 7301.



NOTE:
Center of sensitive volume of tube and centers of lamps lie in plane of paper.

FIGURE 2. Position of tube for photosensitivity test.

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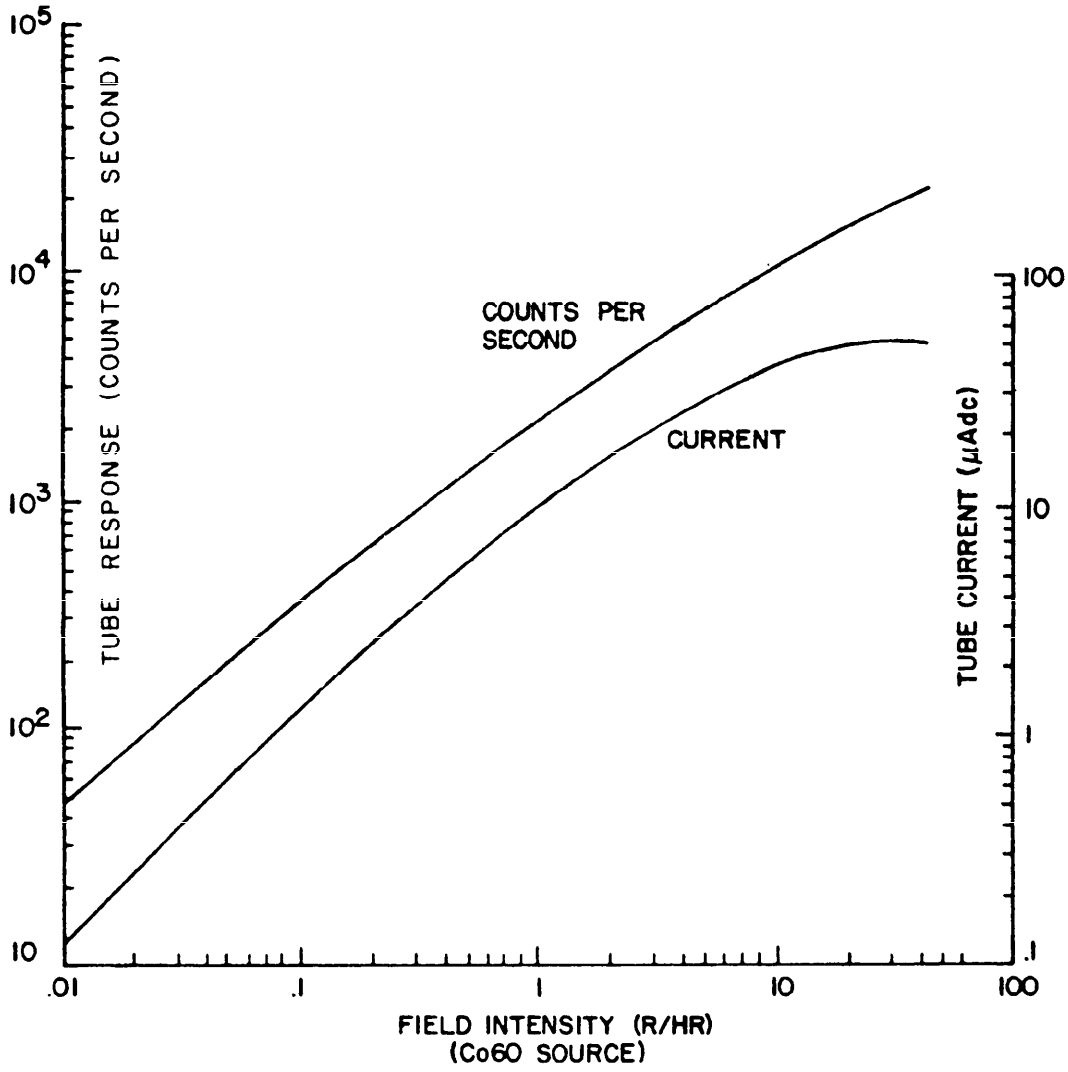


FIGURE 3. Response characteristics of electron tube type 7301.

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